



GENOTYPING PROTOCOL

Investigator: Alyssa Hasty

Genome Edit: Hc Exon 3 deletion

Allele name: *Hc^{em1}* (ILAR lab code)

Hc Exon3 5'Fwd1 primer: CATCATGACATTAGACAATGAAGGG

Hc Exon3 3'Rev1 primer: CACCACCAACCTTCAAAGAGG

Hc WT PCR product = 898 bp

Hc Exon 3 deletion product

- = 288 bp if you breed the -610 bp Hc knockout allele
- = 323 bp if you breed the -575 bp Hc knockout allele
- = 330 bp if you breed the -568 bp Hc knockout allele

PCR Protocol:

PCR reaction (50 ul)

5X NEB Q5 Reaction Buffer	10 ul
10 mM dNTPS	1 ul
10 uM HcExon3-5'Fwd1	2.5 ul
10 uM HcExon3-3'Rev1	2.5 ul
NEB Q5 High-Fidelity DNA Polymerase	0.5 ul
Nuclease free water	32.5 ul
DNA tail lysate	1.0 ul

PCR program

1. 98°C, 30 sec
2. 98°C, 10 sec
3. 56°C, 20 sec
4. 70°C, 45 sec
5. Go to 2, 35 X
6. 70°C, 5 min
7. 4°C, forever